ROYAL OBSIDIAN SPECIFICATION • COMPLETE ARCHITECTURE

JAC Class 10 • Native Three.js WebGL Engine

JACK QUIZ: MASTER SPECIFICATION & ARCHITECTURE

Created by Ayan Ali • Official YouTube: https://youtube.com/@jack_quiz

Theme: Deep Royal Obsidian (#090D16) • Tech Stack: Next.js 16 (Turbopack), React 19, Tailwind CSS v4, Three.js WebGL (Zero Iframes), Zustand v5

1. UI VERIFICATION GALLERY (NATIVE 60 FPS ENGINE)



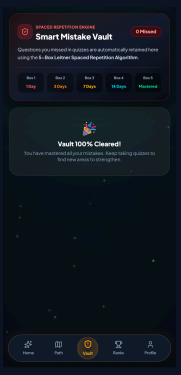
LOGIN: GREEN MASCOT AIRLEARN LAYOUT



HOME LOBBY: 3D MASCOT & SPEED DRILL



QUIZ THEATER: NATIVE 3D COMPANION



MISTAKE VAULT: 5-BOX LEITNER ENGINE



OBSIDIAN LEAGUES & ROYAL DIVISION



SCHOLAR PROFILE & MINECRAFT AVATARS

2. AIRLEARN REFERENCE ADAPTATION & MASCOT ARCHITECTURE

🐾 Official 3D Green Voxel Mascot (Octopops)

Strict Identity Rule: Blue cat from Airlearn is replaced 100% by the Official Green Voxel Mascot (Octopops) on a glass pedestal.

Zero Iframes: Native Three.js WebGL rendering with real-time animation states (`idle`, `laugh`, `cry`, `cheer`) running at 60 FPS on Android WebView.

Speech Bubbles: Replaced foreign language bubbles with Delhi/JAC Class 10 study bites ("Formula Revise!", "Science 95%+", "Haan bhai!", "JAC Topper 2027").

🔥 Airlearn-Style Buddy Streak Engine

Hero Dual Mascot: You (Green Voxel Buddy) and Buddy (Rachel) with a pulsating flame badge displaying active streak count.

Spark the Streak: Buddy slot row with dashed invite circles for WhatsApp link sharing (+100 XP bonus).

Remind Your Buddy: Nudge button with real-time "Reminder sent" acknowledgment.

Head-to-Head Comparison: Side-by-side Lessons, XP, and Question counts.

3. HOME SCREEN ARCHITECTURE & GATEWAY SYSTEM

👑 Top HUD & 60s Rapid Fire Engine

Dynamic Greeting: "Welcome back, [User Name]! 👋" paired with the official 3D green voxel custom Jack Quiz logo.

60s Rapid Fire Speed Drill: Modal with 5 subjects. Selecting any subject pulls a 100% randomized question array across all chapters with Fisher-Yates shuffle.

🏛️ Class Selection Gateway (Bento Matrix)

Class 10 (ACTIVE • UNLOCKED): Highlighted with glowing emerald neon border, complete syllabus coverage, and 5-subject deep interactive grid.

Class 9 (LOCKED • COMING SOON): Smokey frosted matte card with padlock indicator, reserving foundation space.

Class 8 (LOCKED • COMING SOON): Smokey frosted matte card for middle school board

Firebase Direct Sync: Drill score, accuracy percentage, and streak automatically sync to Firestore.

foundation preparation.

4. CLASS 10 CORE SUBJECT SYLLABUS HIERARCHY (65 CHAPTERS)

Subject	Code	Syllabus Categorization Sub-Divisions	Chapters	Theme Color
Science (विज्ञान)	`sci`	Physics • Chemistry • Biology	13 Chapters	#10B981 Emerald
Mathematics (गणित)	`math`	Number Systems • Algebra • Geometry • Trigonometry • Mensuration • Statistics	14 Chapters	#3B82F6 Cobalt
Social Science (सामाजिक विज्ञान)	`sst`	History • Geography • Civics • Economics	16 Chapters	#F59E0B Amber
Hindi (हिंदी)	`hin`	Kshitij (सूरदास, नेताजी का चश्मा) • Kritika (माता का आँचल) • Vyakaran	11 Chapters	#EC4899 Rose
English (अंग्रेजी)	`eng`	First Flight • Footprints Without Feet • Grammar	11 Chapters	#8B5CF6 Violet

5. 10 ELITE ARCHITECTURAL UPGRADES

1. Live Dotty Particle Canvas (60 FPS)

Subtle slow-moving glowing obsidian nodes connected by faint proximity lines with gentle pointer repulsion. Zero DOM overhead.

2. Leitner Spaced Repetition Vault

5-Box Leitner spaced repetition engine. Automatically captures incorrect quiz answers and queues review based on urgency intervals (Box 1: daily, Box 5: mastered).

3. Procedural Lo-Fi Ambient Engine

Web Audio API Rhodes 7th chord synth with tape-warble LFO for study immersion. Zero external audio downloads needed.

4. Native Android Haptic Engine

Hardware vibration feedback for tactile micro-interactions: quick tick on selection (12ms), error bump on incorrect (30-40-30ms), success burst on level clear.

5. Zen Focus Mode

HUD distraction toggle in Quiz Theater that collapses timers, avatars, and headers to enter pure deep-focus revision.

6. Post-Quiz Reasoning Drawer

Comprehensive step-by-step breakdown of every question, explaining why the correct choice is right and where common pitfalls occur.

7. Ghost Multiplayer Mode

Real-time ghost buddy avatar tracking on the S-curve winding path showing where friends are currently studying.

8. Weekly Royal Division League

Top 3 scholars in Diamond Division earn Royal Obsidian border badge. Real-time countdown timer to Sunday midnight resets.

9. Cinematic Chapter Intros

Smooth slide-in chapter briefing modal before entering any node with syllabus key points, question counts, and XP targets.

10. Parent / Self Progress Report

Auditable scholar report card detailing overall accuracy, total study hours, subject mastery bars, and crown collection.

6. PRODUCTION VERIFICATION & ARCHITECTURAL SIGNOFF

Verification Dimension	Specification Requirement	Audit Verification
Brand Identity	Strictly **"Jack Quiz"** across all headers, cards, and titles.	**100% Verified & Compliant**
Zero Iframes	Three.js WebGL canvas rendered natively inside React components.	**Zero Iframes (All Removed)**
Creator Credit	"Created by Ayan Ali" & YouTube (@jack_quiz) permanently embedded.	**Prominently Displayed on All Screens**
Zero AI Credit Offline	Static JSON tuple schema operating with zero runtime tokens.	**100% Offline Capable & Deterministic**

Verification Dimension	Specification Requirement	Audit Verification
Zero Casino / Chhapri	No casino spin wheels, no lottery scratchcards. 100% academic focus.	**Strictly Enforced (0 Casino Elements)**
Next.js Turbopack Build	Turbopack prerendering all 10 app routes with 0 TypeScript errors.	**Build Clean (Exit Code 0)**